DALAS Project
Drug Repurposing

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Contents

1 Introduction

1.1 Problem Statement

Clearly define the problem your project addresses. Explain its relevance in the scientific or practical context.

The discovery of new drugs is, although crucial, an extremely expensive and time-consuming process. Many candidate drugs ultimately turn out to be not only ineffective, but also potentially toxic. It is thus of interest to repurpose already approved drugs for diseases they have not yet been tested against. To prioritize potential clinical trials, in addition to experts' knowledge, it's important to be able to predict the likelihood of success for such studies. This project aims to explore the data and develop a potential prediction pipeline for this purpose. We acknowledge that industrial and academic drug development employ far more comprehensive methods, this project, by contrast, focuses on a preliminary, data-driven approach and is exploratory in nature.

1.2 Motivation and Objectives

Explain why this problem is important. State your main objectives and hypotheses.

The ability to predict the success of drug repurposing has academic and clinical value, it helps reducing the experimental cost and accelerating the discovery of effective treatments. The main objectives of this project are to:

- Explore public datasets to identify some relevant features
- Develop a preliminary machine learning pipeline for success prediction
- Evaluate the model performance
- Identify biases and potential ways for improvement

We decided to specify a group of diverse but closely related diseases - immune system disorders and drugs that are approved for at least one member of this group.

1.3 Pipeline Overview

To address the drug repurposing prediction problem, we developed a data pipeline that integrates multiple public biomedical databases, extracts relevant features, and trains machine learning models. Figures 1 and 2 provide two views of this pipeline.

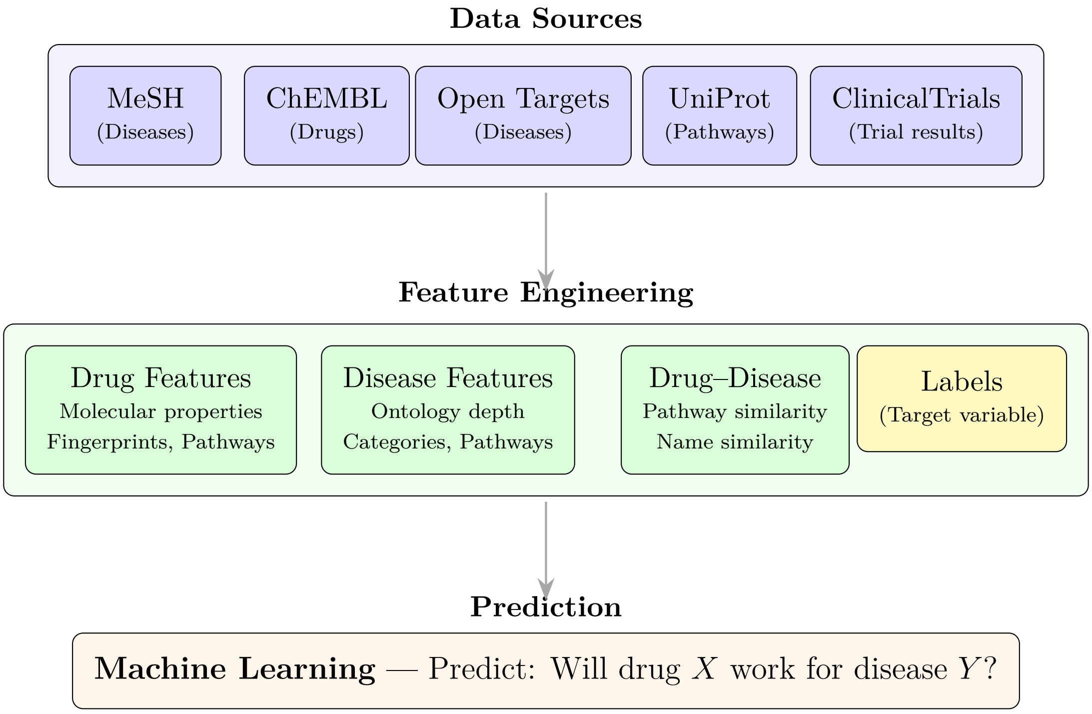

Figure 1: Simplified pipeline overview showing the three main stages of our drug repurposing prediction system. Data from multiple biomedical databases is processed into numerical features, which are then used to train machine learning models. The Labels box (yellow) represents the target variable—trial success or failure—derived from clinical trial outcomes using heuristic rules we defined.

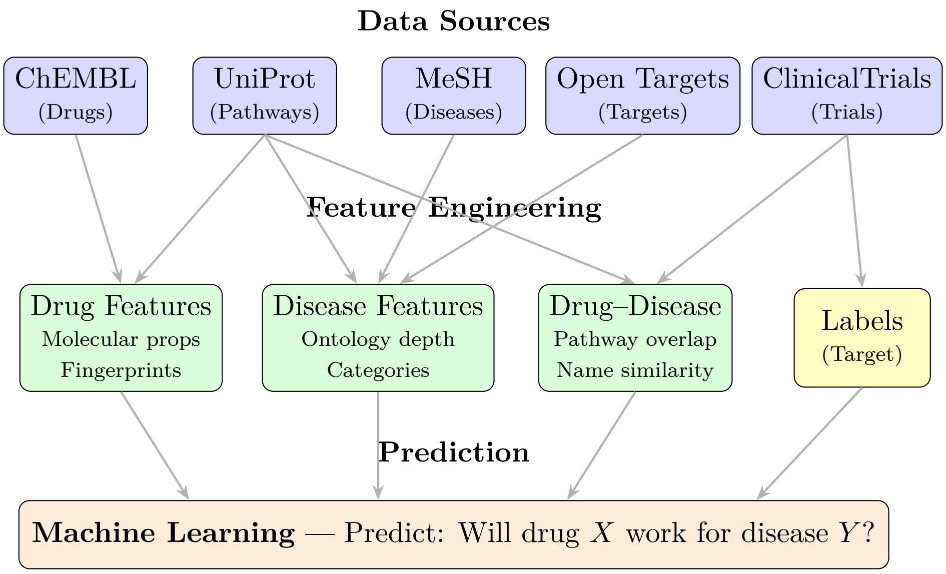

Figure 2: Detailed data flow with reorganized layout. UniProt provides pathway mappings for drugs, diseases, and their similarity. ClinicalTrials provides both trial dates (for Drug–Disease) and outcomes (for Labels).

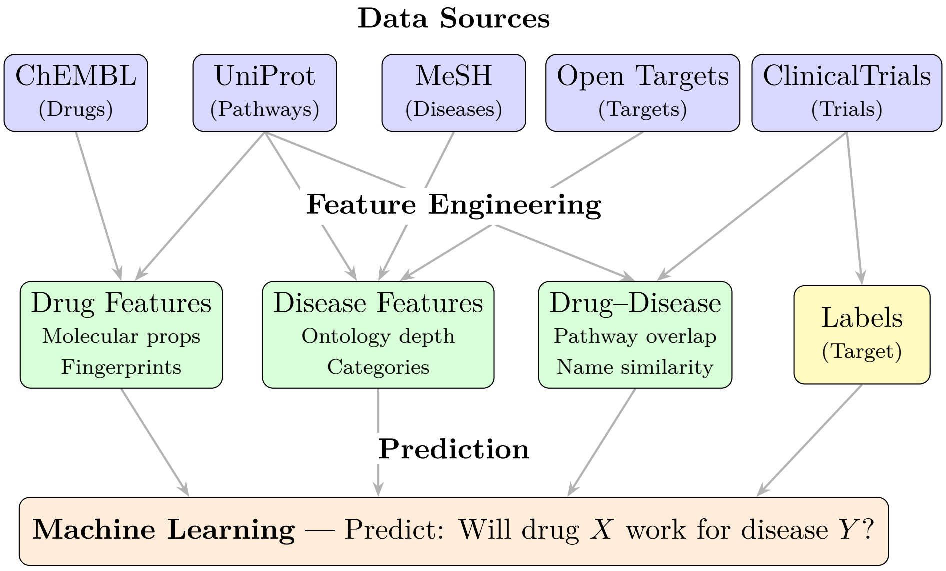

Figure 3: Same as Figure 2 but with labels having white backgrounds to create visual "gaps" where arrows pass behind text.

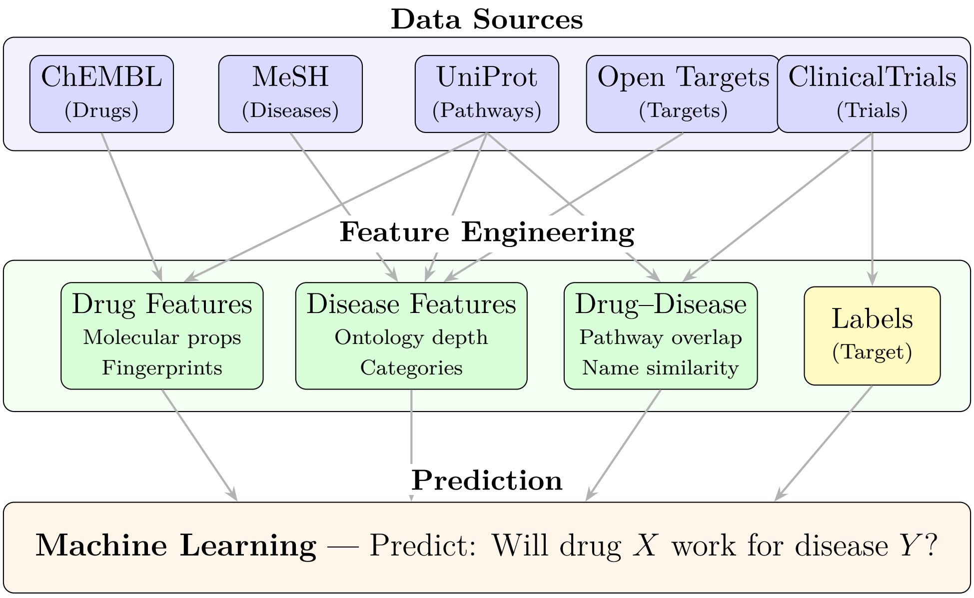

Figure 4: Detailed pipeline showing data flow from sources to features. Arrows indicate which databases contribute to each feature category. UniProt provides pathway mappings used across drug, disease, and pair features. ClinicalTrials provides both trial metadata and outcome labels.

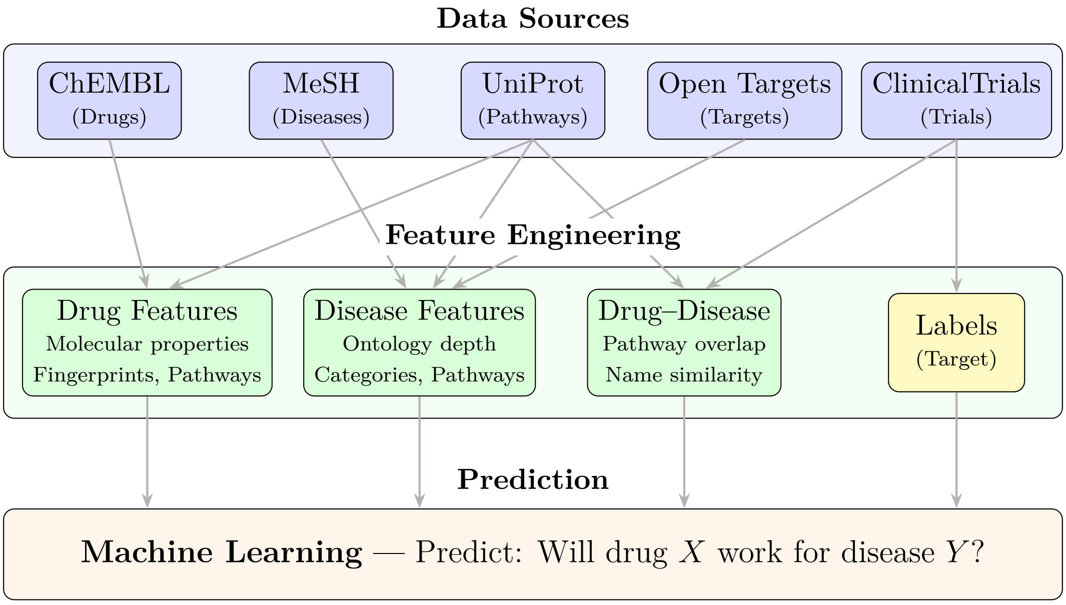

Figure 5: Complete pipeline overview for drug repurposing prediction. Five public biomedical databases provide the raw data, which is transformed into three feature categories plus labels. The yellow Labels box is distinguished because it represents the target variable derived from clinical trial outcomes using heuristic rules, unlike the objectively measured features.

The pipeline consists of three main stages:

1. **Data Collection**: We retrieve disease identifiers from MeSH, drug properties and indications from ChEMBL, disease–target associations from Open Targets, clinical trial outcomes from ClinicalTrials.gov, and protein pathway mappings from UniProt/Reactome.

2. **Feature Engineering**: Raw data is processed into numerical features. Drug features include molecular descriptors and fingerprints; disease features capture ontological structure and pathway information; drug–disease features measure pathway overlap and semantic similarity.

3. **Prediction**: We train several machine learning models (logistic regression, random forests, gradient boosting, etc.) to predict whether a drug–disease pair will succeed in clinical trials.

The following sections detail each stage: Section 2 describes the data sources and preprocessing, Section 3 presents the machine learning methodology, and Section 4 reports our findings.

2 Data

2.1 Data Sources

Describe where the data comes from, collection methods, and aggregation process.

Among the vast amount of medical, pharmacological and chemical publicly available data, we focused our attention on those being able to provide us information on basic classification of diseases (MeSH), drug chemical descriptors and targets (ChEMBL), disease targets (Open Targets) as well as information on drug-disease indication studies (ClinicalTrials). Table 1 summarizes information on data sources.

Table 1: Description of data sources.

Source	**Access Method**	**Data Retrieved**
MeSH RDF	SPARQL endpoint	Disease IDs and ontology
ChEMBL	REST API (Python client with query-style filters)	Drug chemical properties, targets and indications
UniProt	REST API (Python client)	Map target and pathway IDs
Open Targets	GraphQL API	Disease targets
ClinicalTrials.gov	REST API	Trial results and metadata

2.2 Target variable definition

Start with explaining how you got the labels.

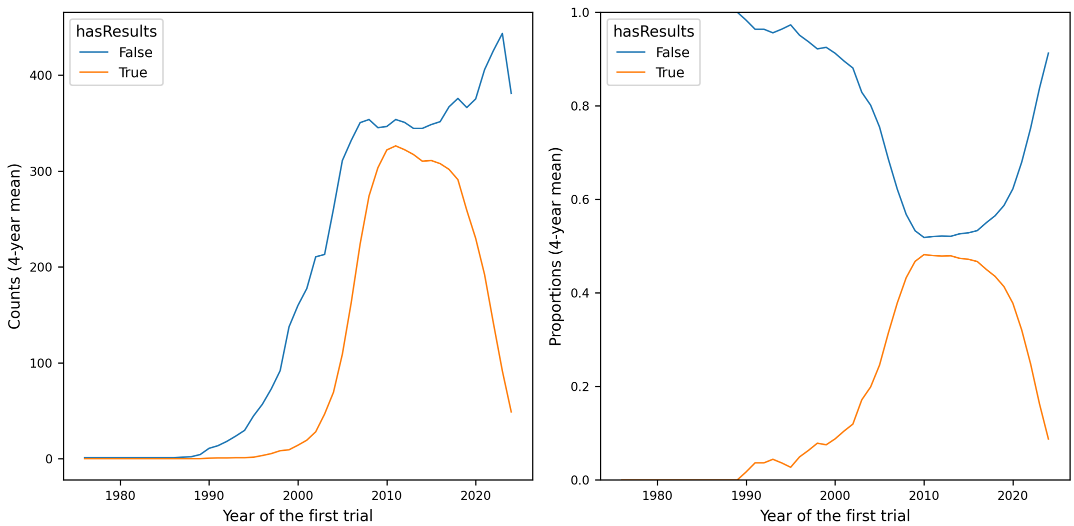

Figure 6: Counts and proportions of published results over the time.

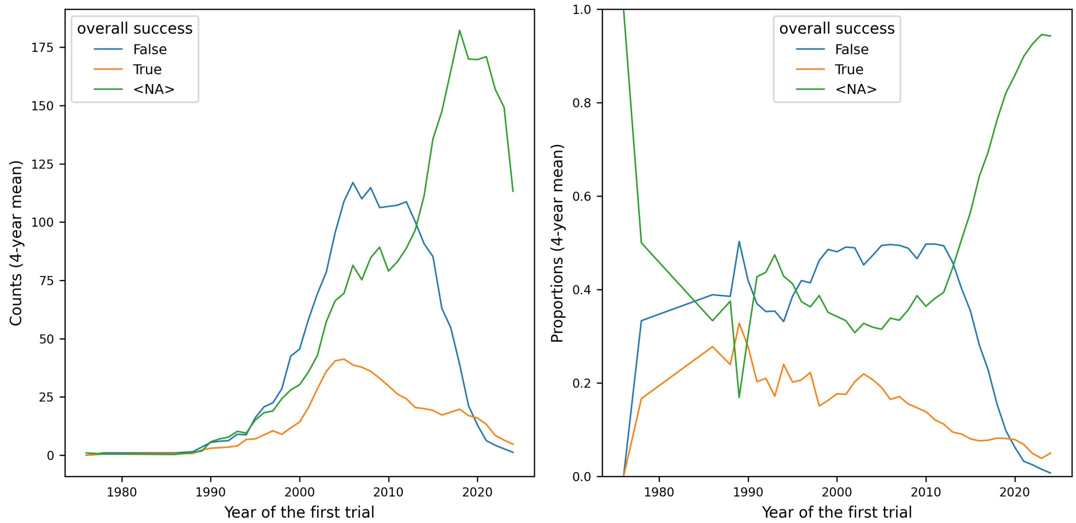

Figure 7: Counts and proportions of label modalities over the time.

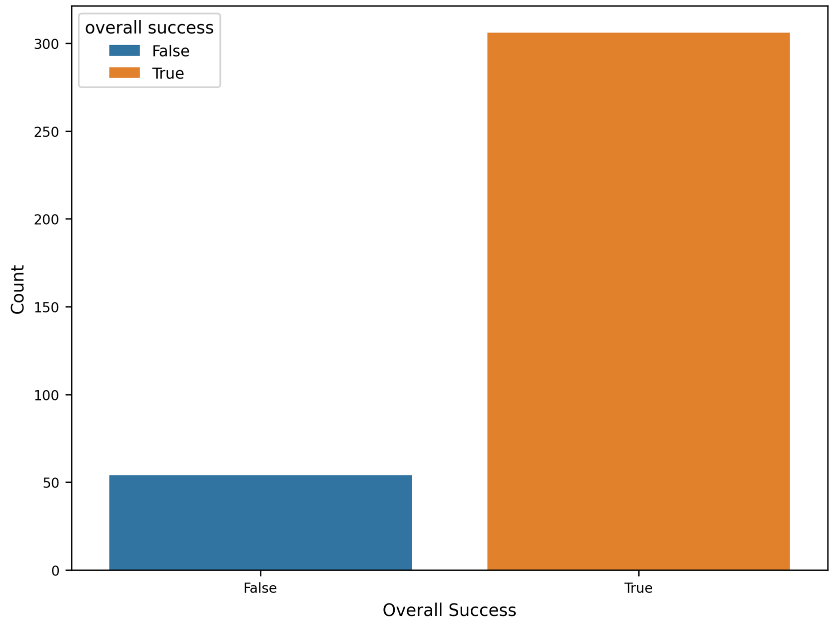

Figure 8: Label distribution for indications with missing date.

2.3 Data Quality and Preprocessing

Discuss missing values, outliers, normalization, feature encoding, or transformations.

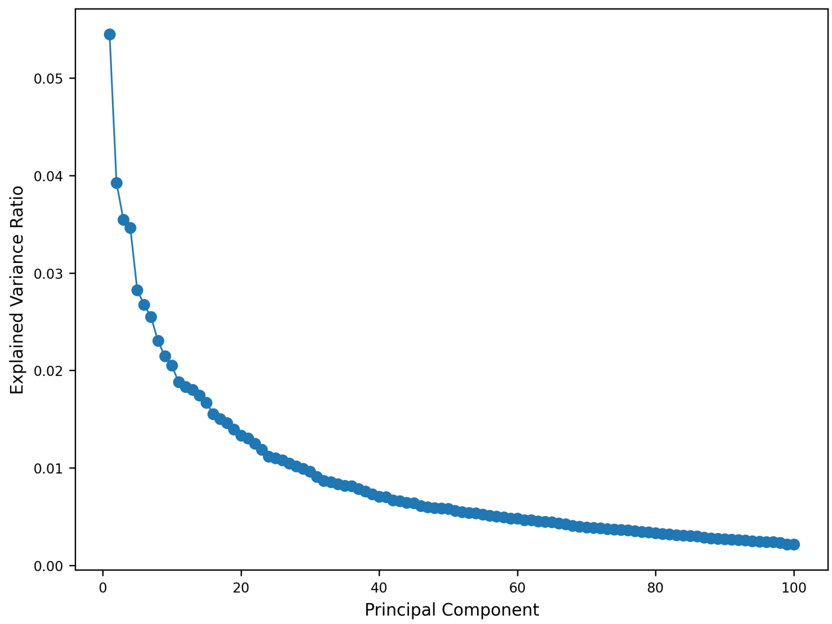

Figure 9: Scree plot of PCA on merged data.

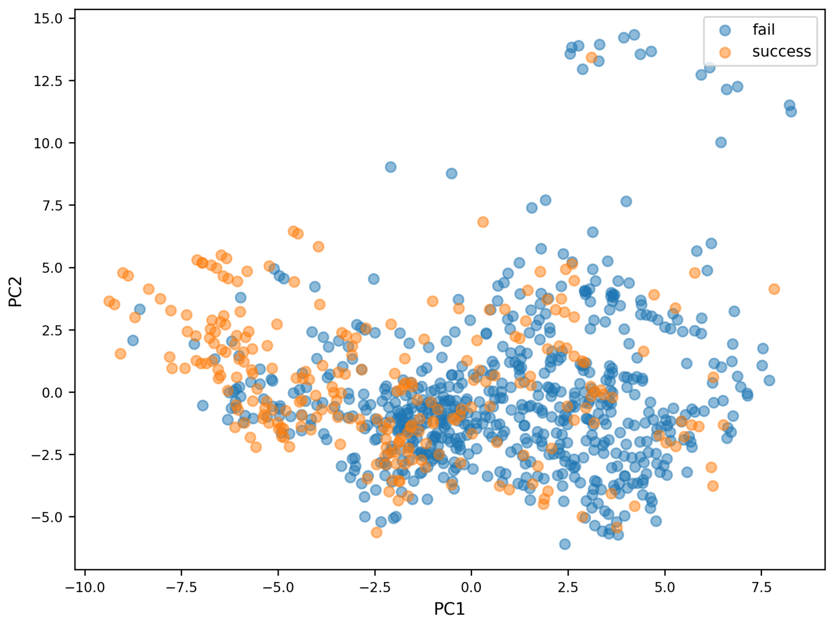

Figure 10: PCA scatter plot.

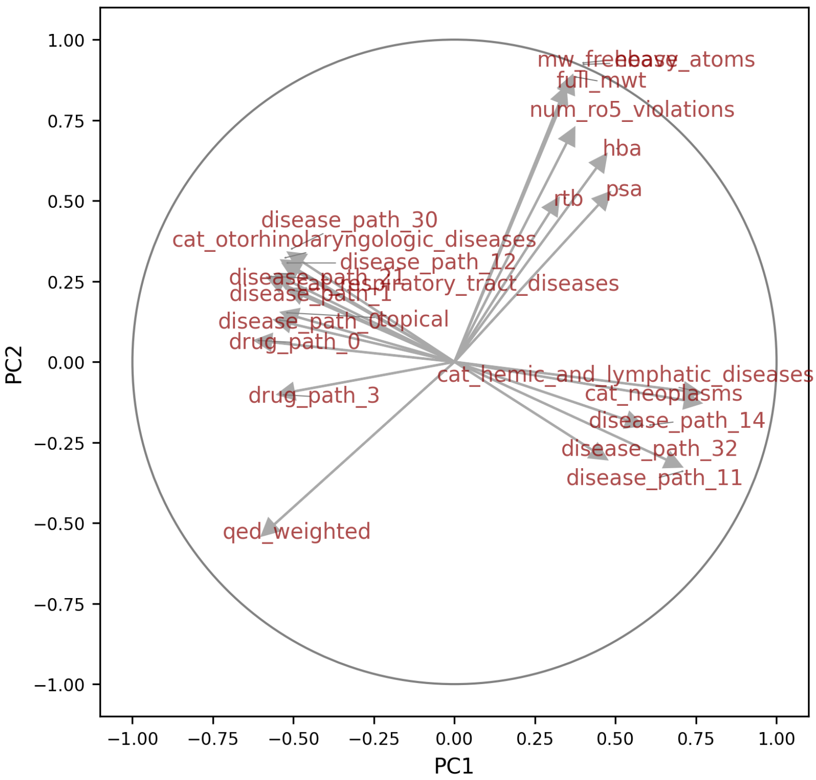

Figure 11: PCA correlation circle.

2.4 Feature Description

Provide a table or list explaining each feature and its meaning. Optionally, include relationships between features.

Table 2: Summary of datasets and merged data.

Dataset	Rows	Columns
Drugs	525	181
Diseases	149	79
Trials	14316	12
Indications	2482	12
Name embeddings	388752	5
Molecular fingerprints	1029	101
Merged data	890	252
Merged data success counts	False: 637, True: 253	

Table 3: Description of features.

Feature	Type	Description
Drug		
drug_id	str	Unique ChEMBL identifier
chemical_probe	bool	Well-characterized and selective, typically used in biological research
dosed_ingredient	bool	Active ingredient that has been dosed in humans
inorganic_flag	bool	Inorganic molecule
natural_product	bool	Derived from a natural source (not fully synthetic)
oral	bool	Known to be administered orally
orphan	bool	Intended for use against a rare condition
parenteral	bool	Known to be administered parenterally
prodrug	bool	Prodrug, activated after administration
therapeutic_flag	bool	Has a therapeutic application, opposed to e.g. imaging
topical	bool	Known to be administered topically
veterinary	bool	Has veterinary application as well
chirality	cat	achiral,single enantiomer, mixture,nan
alogp	float	Octanol-water partition coefficient, lipophilicity indicator
aromatic_rings	int	Number of aromatic rings
full_mwt	float	Molecular weight of the full compound including any salts
hba	float	Number of hydrogen bond acceptors
hbd	float	Number of hydrogen bond donors
heavy_atoms	int	Number of heavy (non-hydrogen) atoms
mw_freebase	float	Molecular weight of parent compound
np_likeness_score	float	Natural product likeness score [Ertl et al., 2008]
num_ro5_violations	int	Violations of Lipinski's rule-of-five, using HBA and HBD definitions
psa	float	Polar surface area
qed_weighted	float	Weighted quantitative estimate of drug likeness [Bickerton et al., 2012]
ro3_pass	int	Passes the rule-of-three (mw < 300, logP < 3 etc)
rtb	int	Number of rotatable bonds
FP_#	float	First PCs in Morgan molecular fingerprint space
drug_path_#	float	First PCs in TF-IDF space derived from pathway lists
Disease		
disease_id	str	Unique EFO identifier
ontology_depth	int	Depth in disease classification tree
nb_descendants	int	Number of descendants in disease classifications tree
nb_indications	int	Number of drug indications
cat_#	bool	Ontological category other than immune system disorders
disease_path_#	float	First PCs in TF-IDF space derived from pathway lists
Drug–Disease		
success	bool	Consensus success or failure
path_similarity	float	Jaccard similarity of pathway lists
name_similarity	float	Cosine similarity of name embeddings in PubMedBERT (later excluded due to data leak)

```
  HeteroData(
drug={ x=[742, 127] },
disease={ x=[167, 25] },
pathway={ x=[1509, 1] },
(drug, interacts, pathway)={ edge_index=[2, 31266] },
```

```
  (disease, associated_with, pathway)={ edge_index=[2, 13608] },
  (drug, tryed_against, disease)={
    edge_index=[2, 1298],
    edge_label=[1298],
  }
)
```

3 Methods

3.1 Model Overview

Briefly describe the models used. Include general principles (e.g., linear models, random forests, neural networks).

3.2 Implementation Details

Mention important parameters, software/libraries used, and cross-validation setup.

4 Results

4.1 Model Comparison

Compare models quantitatively (accuracy, RMSE, AUC, etc.) and qualitatively (strengths/weaknesses).

Table 4: Model comparison summary.

	Model	Mean F1 score	Std F1 score
10	XGBoost	0.642752	0.055461
1	Logistic Regression	0.624640	0.030290
6	Support Vector Machine	0.619498	0.034367
9	Gradient Boost	0.614049	0.068283
8	Random Forest	0.591510	0.035534
0	k-Nearest Neighbors	0.581064	0.064035

Table 5: Best parameters.

	Model	Best parameters
10	XGBoost	colsample_bytree = 0.8,gamma = 0,learning_rate = 0.1,max_depth = 5,n_estimators = 100,subsample = 1.0
1	Logistic Regression	C = 0.1,l1_ratio = 0
6	Support Vector Machine	C = 1,degree = 2,gamma = scale,kernel = poly
9	Gradient Boost	learning_rate = 0.1,max_depth = 5,n_estimators = 300,subsample = 0.8
8	Random Forest	max_depth = 10,min_samples_leaf = 2,min_samples_split = 2,n_estimators = 300
0	k-Nearest Neighbors	n_neighbors = 5,p = 2,weights = distance

4.2 Feature Analysis

Analyze the impact of key features on model performance (e.g., feature importance, correlation).

4.3 Failure and Success Cases

Highlight scenarios where the models perform well or poorly. Include illustrative figures or tables.

5 Discussion

Interpret the results. Relate findings to your original objectives. Discuss limitations and possible improvements.

6 Conclusion

Summarize your work, key findings, and potential future directions.

References

[Bickerton et al., 2012] Bickerton, G. R., Paolini, G. V., Besnard, J., Muresan, S., and Hopkins, A. L. (2012). Quantifying the chemical beauty of drugs. *Nat Chem*, 4(2):90–98.

[Ertl et al., 2008] Ertl, P., Roggo, S., and Schuffenhauer, A. (2008). Natural product-likeness score and its application for prioritization of compound libraries. *J. Chem. Inf. Model.*, 48(1):68–74. Epub 2007 Nov 23.